

PART 5

Testing, UI & Future Enhancements

FinVault – MERN Stack Virtual Stock Trading Platform

DOCUMENT INFORMATION

Project Title	FinVault – Virtual Stock Trading Platform
Document Phase	Phase 1: Project Documentation
Part	Part 5 – Testing, UI & Future Enhancements
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Institution	G. Pullaiah College of Engineering and Technology (GPCET)
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Status	Final Version

1. Testing Methodology

Quality assurance and validation were carried out systematically across all features.

1.1 API Testing with Thunder Client

Thunder Client was used to test endpoint responses, check status codes, verify headers, and audit auth rules. Specific scenarios validated include:

- **Registration Validation:** Registering duplicate emails fails with correct HTTP error codes.
- **Security Protection:** Accessing protected `/api/portfolio` without a Bearer JWT returns 401 Unauthorized.
- **Admin Isolation:** Requesting `/api/admin/users` using a standard user JWT returns 403 Forbidden.
- **Trading Boundaries:** Attempting to purchase stocks costing more than the wallet balance returns a 400 Bad Request indicating insufficient funds.

1.2 Database Integrity Tests

Verified that MongoDB transactions run atomically. For example, stock purchases fail if user balance updates succeed but stock quantity updates fail, ensuring database consistency.

2. User Interface Design

The frontend UI is styled with Vanilla CSS to keep it lightweight, fast, and fully responsive.

2.1 Navigation & Layout

A dark left navigation bar provides quick access to major views, while the right side displays the main page content.

2.2 Primary Pages

- **Landing Page:** Welcomes users, outlining features and how mock stock trading works.
- **Dashboard:** Central hub containing stock search, live listings, and mini-cards showing wallet assets.
- **Portfolio page:** Dynamic grid displaying holding assets, average cost, current pricing, and net ROI.
- **Watchlist Page:** Personalized pinned stock cards showing price variations.
- **Transactions Log:** Ledger showing histories of BUY and SELL activities.
- **Admin View:** Console with metrics and CRUD options for stock assets.

3. Future Enhancements

Plans for expanding FinVault include:

- **Real-time Price Integration:** Replacing mock database updates with live WebSockets or external financial APIs (e.g., Alpha Vantage, Yahoo Finance).
- **AI Advisory Engine:** Integrating a Gemini-powered chatbot to analyze portfolio performance and offer mock stock recommendations.
- **Advanced Account Security:** Adding email authentication, password reset flows, and Two-Factor Authentication (2FA).
- **Report Export Tools:** Enabling users to download trading history statements as Excel sheets or formatted PDFs.
- **Mobile Port:** Building a React Native mobile app for Android/iOS.

4. Academic & Course Metadata

This project is developed as a course requirement for Computer Science & Engineering degree courses.

Field	Details
Developer	Ganesh Basode
Department	Department of Computer Science and Engineering
Institution	G. Pullaiah College of Engineering and Technology (GPCET)
Project Domain	Full Stack Web Application Development (MERN)
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